

$$\frac{A_t - A_\infty}{A_0 - A_\infty} = \gamma(t) \exp\{-k_1[\text{CH}_3\text{SH}]t\} \quad (\text{e})$$

where

$$\gamma(t) = [\epsilon\{\text{NO}_2\} - \beta_t\epsilon\{\text{CH}_3\text{SNO}\} + \epsilon\{\text{CH}_3\text{SNO}\} \times (\beta_t - \beta_\infty) \exp\{k_1[\text{CH}_3\text{SH}]t\}] / [\epsilon\{\text{NO}_2\} - \beta_\infty\epsilon\{\text{CH}_3\text{SNO}\}]$$

where $\epsilon\{i\}$ is the extinction coefficient for species i and

$$\beta_t = [\text{CH}_3\text{SNO}]_t / ([\text{NO}_2]_0 - [\text{NO}_2]_t)$$

$\gamma(t)$ must have values between 1.0 and $\approx 1 + (\beta_0 - \beta_\infty)\epsilon\{\text{CH}_3\text{SNO}\}/\epsilon\{\text{NO}_2\}$. From our earlier data with excess CH_3SH , $\beta_\infty \approx 2/3$,⁶ whereas the above mechanism predicts $\beta_0 = 1.0$. Our results indicate (see Figure 1) that the residual absorption from the CH_3SNO is only $\approx 5\%$ of that of the original NO_2 . Therefore $\beta_\infty\epsilon\{\text{CH}_3\text{SNO}\}/\epsilon\{\text{NO}_2\} \approx 0.05$, and $\gamma(t)$ probably never exceeds ≈ 1.02 . For all practical purposes $\beta(t)$ is constant, as confirmed by the linearity of our plots of $\ln\{(A_t - A_\infty)/(A_0 - A_\infty)\}$ vs. reaction time. Thus $\beta(t)$ can be taken as 1.00 throughout any run.

The mechanism predicts that

$$-\ln\{(A_t - A_\infty)/(A_0 - A_\infty)\} = k_1[\text{CH}_3\text{SH}]t \quad (\text{f})$$

However, we have found that the observed rate law was

$$-\ln\{(A_t - A_\infty)/(A_0 - A_\infty)\} = (k_0 + k_1[\text{CH}_3\text{SH}])t \quad (\text{g})$$

Thus k_0 represents an additional first-order NO_2 -loss term not

involving CH_3SH . Since this term is larger in the 3-L cell than in the 12-L cell for comparable temperatures, k_0 presumably represents a wall-loss term.

The Arrhenius plot for k_1 based on a single least-squares straight line gives a preexponential factor of $1.6 \times 10^{-13} \text{ cm}^3/\text{s}$ and an activation energy of 12.6 kcal/mol. The preexponential factor is too large to represent a four-center reaction such as implied by reaction 1a. Therefore if reaction 1a is occurring with a preexponential factor of $1.6 \times 10^{-13} \text{ cm}^3/\text{s}$, it must proceed through an activated complex which is a two-center addition reaction. Thus if reaction 1 is a single reaction, the activated complex corresponds to addition of the NO_2 to the sulfur in CH_3SH , regardless of whether reaction 1a or 1b is occurring.

On the other hand, the Arrhenius plot is better fitted by the sum of two Arrhenius expressions. Perhaps one of these corresponds to reaction 1a and the other to reaction 1b. One of these rate coefficients is $3.2 \times 10^{-16} \exp\{-4300/T\} \text{ cm}^3/\text{s}$ and presumably corresponds to reaction 1a, because the preexponential factor is small enough to correspond to a four-center reaction. The other rate coefficient is $2.0 \times 10^{-9} \exp\{-10300/T\} \text{ cm}^3/\text{s}$. The preexponential factor exceeds collision frequency and cannot correspond to a simple bimolecular reaction. Presumably it represents some unknown long-chain mechanism initiated by reaction 1b.

Acknowledgment. The authors thank Dr. R. Simonaitis for helpful discussions.

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Predictions of Pressure and Temperature Effects upon Radical Addition and Recombination Reactions

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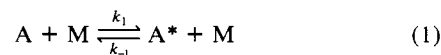
Accurate kinetic models of high-temperature hydrocarbon systems require detailed knowledge of the temperature and pressure dependence for both addition and recombination reactions. Since both proceed via a long-lived collision complex, they can be treated as chemical activation processes. A method is described which combines chemical activation distribution functions and QRRK methods to predict dissociation rates to estimate the kinetics of these systems. Generic Arrhenius preexponential factors for decomposition of the energized complexes have been obtained by analysis of a number of addition and recombination reactions and the appropriate equilibrium constants. With those values, it becomes a simple exercise to calculate the rate constants as functions of temperature and pressure. Comparisons of those calculations to the limited literature data suggest this approach provides reliable rate constant estimates. Several specific applications are considered. These include $\text{CH}_3 + \text{CH}_3$ recombination, CH_3 and C_2H addition to unsaturates, and isomerization of the energized complex. Those results are also compared to several master equation calculations which consider a more realistic case for collisional energy transfer; the results suggest the simpler approach can often provide reliable results. The results identify previously overlooked high-temperature reaction channels that could profoundly affect hydrocarbon kinetics by allowing rapid conversion of relatively unreactive hydrocarbon radicals to highly reactive hydrogen atoms.

Introduction

In recent years, kineticists have become increasingly concerned about the effect of temperature on the falloff behavior of dissociation-recombination reactions. Warnatz¹ has explicitly calculated falloff curves for some key dissociation reactions as functions of temperature. The influence of temperature can be dramatic: increasing the temperature from 300 to 1350 K for $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$ shifts the falloff curve by 4 orders of magnitude toward higher pressures. The unnerving consequence of such calculations is that, for many molecules of interest, explicit attention must be paid to falloff effects. It is clearly not adequate to continue to

assume that most dissociation reactions are at their high-pressure limit at high temperatures. This point is also evident in the recent work by Larson et al.² where parameters are presented for estimation of the falloff behavior of several molecules of interest to combustion modelers.

Qualitative reasons for the shift of falloff curves to higher pressure as the temperature is increased can be found in the simplified Lindemann-Hinshelwood model:³



(1) Warnatz, J. *Combust. Sci. Technol.* **1983**, 34, 177.

(2) Larson, C. W.; Patrick, R.; Golden, D. M. *Combust. Flame* **1984**, 58, 229.

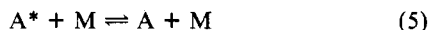


Employing the steady-state assumption for the energized molecule A^* , we have

$$k_{\text{uni}} = \frac{k_2 k_1 [M]}{k_{-1} [M] + k_2} \quad (3)$$

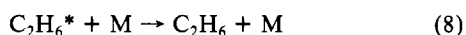
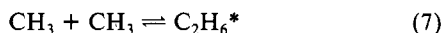
The high-pressure limit ($k_{\text{uni}} \neq f([M])$) is characterized by $k_{-1} [M] \gg k_2$. Since k_2 increases with temperature much faster than k_{-1} , higher pressures are required to maintain the inequality.

Precisely the same behavior is expected for addition reactions since they follow an analogous mechanism



where A^* represents the energized adduct. Quantitative treatments of such additions were pioneered by the chemical activation work of Rabinovitch and co-workers.⁴ Most of this work involved addition of H atoms ($R = H$) and the primary emphasis was elucidation of information on vibrational energy transfer. This approach has been extended to analyze the temperature and pressure dependence for addition reactions involving OH ^{5,6} and insertions of CH .⁷ However, additions involving hydrocarbon radicals such as CH_3 are still often assumed to be at the high-pressure limit, even at elevated temperatures.⁸

Calculations presented in this paper suggest that many hydrocarbon addition reactions are *not* in their high-pressure limit at elevated temperatures, even at pressure of 1 atm or above. Moreover, the same rapid rise in internal energy with temperature, which shifts these reactions away from the high-pressure limit, also allows access to new endothermic reaction channels, further complicating the situation. Similar endothermic channels can also open up in recombination reactions. One such case involves methyl radicals⁹⁻¹¹



Here a chain termination event can be transformed into chain propagation in which CH_3 is replaced by the much more reactive H atom. Similar recombination reactions involving radicals containing nitrogen have been proposed to explain the kinetics of ammonia flames.^{12,13} Our present calculations suggest similar dramatic effects for addition reactions, where relatively unreactive hydrocarbon radicals can be replaced with H atoms.

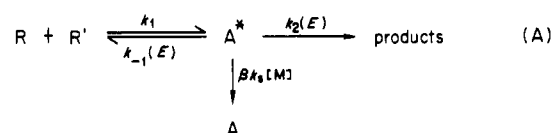
A straightforward method is proposed to estimate the magnitude of these effects over extended temperature and pressure ranges. The procedure used is applicable to both recombination and ad-

dition reactions. Comparison to recombination data indicates the method can provide estimates of sufficient accuracy to be useful for modeling calculations. A particularly attractive feature of this approach is that the requisite input for the calculations is easy to develop.

The next section outlines the basis for the calculations. This is followed by a discussion of the methods used to develop the necessary parameters for use in specific applications. Several specific examples illustrating various applications are then presented.

Development of the Working Equations

The first case considers recombination or addition to occur via the sequence:



Here R is a radical, R' is a radical (recombination) or unsaturated molecule (addition), A^* is the energized complex which can either dissociate or be collisionally stabilized, β is the collisional deactivation efficiency, and k_s is the collisional rate constant for stabilization. The dissociation rate constants k_{-1} and k_2 are written as explicit functions of energy. These energy dependencies are best treated via RRKM theory.¹⁴ However, a simpler approach, quantum RRK theory (QRRK), will be used here. The major advantage of QRRK is a substantial simplification of the input data required; one only needs to estimate a geometric mean frequency for the complex, rather than consider each specific frequency. Moreover, such a simplification still permits reasonable predictions of falloff curves.^{1,14}

Applying the steady-state assumption to A^* , one can define two apparent rate constants, a stabilization rate constant k_{stab}

$$k_{\text{stab}} = \frac{d[A]/dt}{[R][R']} = k_1 \sum_{E_{\text{crit}}} \frac{\beta k_s [M]}{\beta k_s [M] + k_{-1}(E) + k_2(E)} f(E) \quad (10)$$

and the rate constant k_{rxn} for formation of other dissociation products from A^*

$$k_{\text{rxn}} = \frac{d[\text{products}]/dt}{[R][R']} = k_1 \sum_{E_{\text{crit}}} \frac{k_2(E)}{\beta k_s [M] + k_{-1}(E) + k_2(E)} f(E) \quad (11)$$

where $f(E)$ is the fraction of A^* at a specific energy E , the sums are in steps of $h\nu$ (ν = geometric mean frequency), and E_{crit} is the threshold energy for dissociation to reactants. The chemical activation distribution function, $f(E)$, can be expressed as¹⁵

$$f(E) = \frac{k_{-1}(E)K(E)}{\sum_{E_{\text{crit}}} k_{-1}(E)K(E)} \quad (12)$$

with

$$K(E) = \alpha^n (1 - \alpha)^s \frac{(n + s - 1)!}{n!(s - 1)!} \quad (13)$$

where $\alpha = \exp(-h\nu/kT)$, s = number of vibrational degrees of freedom, and $n = E/h\nu$, i.e., the number of quanta of energy. $K(E)$ is the ratio of excitation to deexcitation rate constants for the quantum version of Kassel's theory using Hinshelwood's concept of excitation with many degrees of freedom.¹⁶ The energy-dependent rate constants can be written as

$$k_i(E) = A_i \frac{n!(n - m_i + s - 1)!}{(n - m_i)!(n + s - 1)!} \quad (14)$$

(14) Robinson, P. J.; Holbrook, K. A. "Unimolecular Reactions"; Wiley-Interscience: London, 1972.

(15) Reference 14, p 272 ff.

(16) Reference 14, p 54.

(3) Laidler, K. J. "Theories of Chemical Reaction Rates"; McGraw-Hill: New York, 1969.

(4) Tardy, D. J.; Rabinovitch, B. S. *Chem. Rev.* **1977**, *77*, 369 and references therein.

(5) Bartels, M.; Hoyermann, K.; Sievert, R. *Symp. (Int.) Combust., [Proc.]*, **19**, 1982 **1983**, 61.

(6) Smith, G. P.; Fairchild, P. W.; Crosley, D. R. *J. Chem. Phys.* **1984**, *81*, 2667.

(7) Berman, M. R.; Lin, M. C. *Chem. Phys.* **1983**, *82*, 435.

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(9) Roth, P.; Just, T. *Ber. Bunsenges. Phys. Chem.* **1979**, *83*, 577.

(10) Kiefer, J. H.; Budach, K. A. *Int. J. Chem. Kinet.* **1984**, *16*, 679.

(11) Larson, C. W.; Golden, D. M., presented at the Western States Section—Combustion Institute, 1984; Paper No. 84-91.

(12) Miller, J. A.; Smooke, M. D.; Green, R. M.; Kee, R. J. *Combust. Sci. Technol.* **1983**, *34*, 149.

(13) Dean, A. M.; Chou, M. S.; Stern, D. *Int. J. Chem. Kinet.* **1984**, *16*, 633.

where A_i is the high-pressure Arrhenius preexponential factor for reaction i , and m_i is the number of quanta ($E_i/h\nu$) corresponding to the energy threshold for reaction i .

The values of the collisional stabilization rate constant (k_s) are computed by using Lennard-Jones collision rates¹⁷

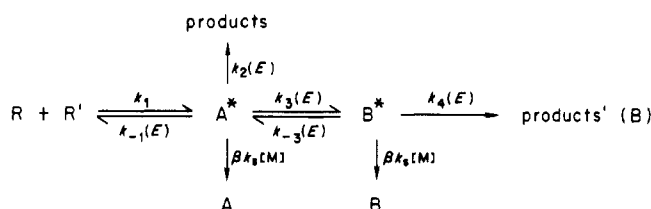
$$k_s = 2.708 k_{\text{HS}} (\epsilon/kT)^{1/3} \quad (15)$$

where k_{HS} is the usual hard-sphere collision rate constant. Values for collision diameters and well depths were taken from Kee et al.¹⁸ Collision efficiencies (β) were evaluated from Troe's¹⁹ expression:

$$\frac{\beta}{1 - \beta^{1/2}} \cong \frac{-\langle \Delta E \rangle}{F_E kT} \quad (16)$$

For this work, F_E was assumed to be 1.15 (cf. Table II of ref 19). Values of the average energy transferred ($-\langle \Delta E \rangle$) were taken from Gardiner and Troe.²⁰

Treatment of Isomerization of the Energized Complex. The present approach has also been extended to allow the energized complex to isomerize. This can be represented by the scheme:



As before, one can define the appropriate apparent rate constants

$$k_{\text{stab}_A} = \frac{d[A]/dt}{[R][R']} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{\beta k_s[M] D'}{DD' - k_{-3}(E)k_3(E)} f(E) \quad (17)$$

$$k_{\text{rxn}_A} = \frac{d[\text{products}]/dt}{[R][R']} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{k_2(E) D'}{DD' - k_{-3}(E)k_3(E)} f(E) \quad (18)$$

$$k_{\text{stab}_B} = \frac{d[B]/dt}{[R][R']} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{\beta k_s[M] k_3(E)}{DD' - k_{-3}(E)k_3(E)} f(E) \quad (19)$$

$$k_{\text{rxn}_B} = \frac{d[\text{products}']/dt}{[R][R']} = k_1 \sum_{E_{\text{crit}}}^{\infty} \frac{k_4(E) k_3(E)}{DD' - k_{-3}(E)k_3(E)} f(E) \quad (20)$$

where $D = k_{-1}(E) + \beta k_s[M] + k_2(E) + k_3(E)$ and $D' = k_{-3}(E) + \beta k_s[M] + k_4(E)$. The critical number of quanta (m_i in (14)) for $k_{-1}(E)$, $k_2(E)$, and $k_3(E)$ are calculated from the bottom of the potential well of A, while that for $k_{-3}(E)$ and $k_4(E)$ are based on the bottom of the potential well of B.

Treatment of Collisional Deactivation. A problem with use of βk_s in the context of scheme A or B is that values of β were deduced for systems where only one dissociation channel was considered as opposed to the two or more in these schemes. Troe's approach allows one to use a modified strong collision assumption in that every collision is not deactivating, but whenever deactivation does occur it is complete; i.e., all the excess energy is lost. The decrease in β with increasing temperature then allows for the somewhat higher excitation levels with correspondingly lower probability of deactivation below the critical energy. The introduction of a second dissociation channel complicates this issue since it is now possible for A^* to lose sufficient energy so that one channel is not accessible while reaction (at a slower rate) could

TABLE I: Estimated (High Pressure) Rate Constants for C-C Rupture in Hydrocarbon Molecules

reaction	$\log A_i, \text{s}^{-1}$	$E_i, \text{kcal mol}^{-1}$	$\Delta H^\circ_{300}, \text{kcal mol}^{-1}$
A. Nonresonant Dissociation Products^a			
$\text{C}_2\text{H}_6 \rightarrow 2\text{CH}_3$	16.9	89.4	89.8
$\text{C}_3\text{H}_8 \rightarrow \text{C}_2\text{H}_5 + \text{CH}_3$	16.9	84.4	85.5
$n\text{-C}_4\text{H}_{10} \rightarrow \text{C}_3\text{H}_7 + \text{CH}_3$	17.0	84.7	85.9
$n\text{-C}_4\text{H}_{10} \rightarrow 2\text{C}_2\text{H}_5$	16.9	80.2	81.9
$\text{C}=\text{CC} \rightarrow \text{C}=\dot{\text{C}} + \text{CH}_3$	16.9	99.5	100.3
$\text{CC}=\text{CC} \rightarrow \text{CC}=\dot{\text{C}} + \text{CH}_3$	17.5	99.3	100.2
B. One Resonant Product^b			
$\text{C}=\text{CCC} \rightarrow \text{C}=\text{CC} + \text{CH}_3$	15.7	75.8	76.7
$\text{C}=\text{CCC} \rightarrow \text{C}=\text{CC} + \text{CH}_3$	16.0	73.0	73.9
C. Two Resonant Products^c			
$\text{C}=\text{CCCC}=\text{C} \rightarrow 2\text{C}=\text{CC}$	14.8	61.9	63.3
$\text{C}=\text{CCCC}=\text{C} \rightarrow \text{C}=\text{CC} + \text{C}=\text{CC}$	15.1	59.4	60.8
$\text{C}=\text{CCCC}=\text{C} \rightarrow 2\text{C}=\text{CC}$	15.3	56.8	58.2

^a Suggested values: $A_i \approx 1 \times 10^{17} \text{ s}^{-1}$; $E_i \approx \Delta H^\circ_{300} - 1 \text{ kcal mol}^{-1}$. ^b Suggested values: $A_i \approx 8 \times 10^{15} \text{ s}^{-1}$; $E_i \approx \Delta H^\circ_{300} - 1 \text{ kcal mol}^{-1}$. ^c Suggested values: $A_i \approx 1 \times 10^{15} \text{ s}^{-1}$; $E_i \approx \Delta H^\circ_{300} - 1 \text{ kcal mol}^{-1}$.

continue through a lower energy channel. Thus use of βk_s (which "kills" both channels) will introduce a bias in the result when the energy gap between channels is larger than the energy transferred per collision.²¹ For hydrocarbon systems, $-\langle \Delta E \rangle \gtrsim 2 \text{ kcal mol}^{-1}$ ²⁰ and differences in exit channel barriers much greater than this could cause problems.

However, the situation is not as bleak as it first appears. First, the energetics of many CH_3 and C_2H_3 additions are such that the barriers for the two channels are often quite close in energy and the bias introduced should be minimal. For C_2H additions and certain cyclizations, there is a much lower exit channel where one might expect considerable underestimation of the rate through these channels. However, the results shown below indicate these are major routes. Here the barriers are so low that the dissociation rate is extremely fast, faster than deactivation. Thus (fortunately!) the imperfect treatment of energy transfer is often acceptable since collisions are relatively unimportant.

The magnitude of the uncertainty introduced by using a modified strong collision assumption was assessed by explicit comparisons to master equation solutions.²¹ These results are discussed below.

Estimation of Generic Kinetic Parameters. The most important input parameters are the high-pressure Arrhenius preexponential factors A_i and barrier heights E_i for the various reaction channels. Although those could be done on a case-by-case basis, a detailed examination suggested that one could do quite well with a handful of generic estimates. These estimates were obtained by first categorizing the dissociative events in hydrocarbons into four groups: (a) C-C rupture in stable molecules, (b) C-H rupture in stable molecules, (c) C-C rupture in radicals, and (d) C-H rupture in radicals. The reverse rate constants [recombination for (a) and (b), addition for (c) and (d)] were then obtained from the literature. These were combined with linear least-squares fits to the equilibrium constants ($T = 300\text{--}1900 \text{ K}$) for the appropriate reactions to obtain estimates of the dissociation rates. This approach neglects the effects of curvature in the Van't Hoff plots; however, in most cases surprisingly good linear fits were obtained over this extended temperature range.

Results from group a are summarized in Table I. Here the recombination rate was assumed to be $2 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at all temperatures.²² Note the A factors group nicely; use of the suggested values will probably give a result within a factor of 2 of the actual value for any C-C rupture in most stable molecules. The substantial decrease in A when resonantly stabilized radicals are formed is consistent with a lower ΔS for dissociation since

(17) Forst, W. "Theory of Unimolecular Reactions"; Academic Press: New York, 1973; p 184.

(18) Kee, R. J.; Warnatz, J.; Miller, J. A. "A Fortran Computer Code Package for the Evaluation of Gas Phase Viscosities, Conductivities, and Diffusion Coefficients"; Sandia National Laboratories Report SAND 83-8209, 1983.

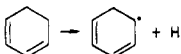
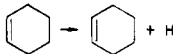
(19) Troe, J. *J. Phys. Chem.* **1979**, *83*, 114.

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(21) Larson, C. W.; Rabinovitch, B. S. *J. Chem. Phys.* **1969**, *51*, 2293.

(22) Allara, D. L.; Shaw, R. *J. Phys. Chem. Ref. Data* **1980**, *9*, 523.

TABLE II: Estimated (High Pressure) Rate Constants for C-H Rupture in Hydrocarbon Molecules

reaction	$\log A_i, s^{-1}$	$\log [A_i/\#H], s^{-1}$	$E_i, \text{kcal mol}^{-1}$	$\Delta H^\circ_{300}, \text{kcal mol}^{-1}$
A. Nonresonant Dissociation Products ^a				
$\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$	16.0	15.4	105.0	104.8
$\text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_5 + \text{H}$	16.1	15.3	98.0	98.2
$\text{C}_3\text{H}_8 \rightarrow i\text{-C}_3\text{H}_7 + \text{H}$	15.8	15.5	94.7	95.1
$\text{C}_3\text{H}_8 \rightarrow n\text{-C}_3\text{H}_7 + \text{H}$	16.2	15.4	97.5	97.9
$n\text{-C}_4\text{H}_{10} \rightarrow n\text{-C}_4\text{H}_9 + \text{H}$	16.2	15.4	98.0	98.2
$n\text{-C}_4\text{H}_{10} \rightarrow s\text{-C}_4\text{H}_9 + \text{H}$	16.0	15.4	95.0	95.2
$\text{C}_2\text{H}_4 \rightarrow \text{C}_2\text{H}_3 + \text{H}$	16.3	15.7	110.0	109.9
$\text{C}=\text{CC}=\text{C} \rightarrow \text{C}=\text{CC}=\text{C}^\cdot + \text{H}$	16.2	15.6	110.0	109.9
B. Resonant Radical Product ^b				
$\text{CC}=\text{CC} \rightarrow \text{CC}=\text{CC}^\cdot + \text{H}$	15.5	14.7	87.3	87.3
$\text{CC}=\text{CC} \rightarrow \text{CC}=\text{CC}^\cdot + \text{H}$	15.6	14.8	85.1	85.1
$\text{C}=\text{C}=\text{CC} \rightarrow \text{C}=\text{C}=\text{CC}^\cdot + \text{H}$	15.3	14.8	85.9	86.0
$\text{C}=\text{CCC} \rightarrow \text{C}=\text{CC}^\cdot\text{C} + \text{H}$	15.1	14.8	82.4	82.5
$\text{C}=\text{CCCC} \rightarrow \text{C}=\text{CCC}^\cdot\text{C} + \text{H}$	15.3	14.7	82.7	83.0
$\text{C}=\text{CCCC} \rightarrow \text{C}=\text{CCC}^\cdot\text{C} + \text{H}$	15.4	14.8	82.4	82.5
	15.7	15.1	72.6	73.2
	15.7	15.1	81.7	81.9

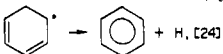
^a Suggested values: $A_i/\#H \approx 3 \times 10^{15} s^{-1}$; $E_i \approx \Delta H^\circ_{300}$. ^b Suggested values: $A_i/\#H \approx 7 \times 10^{14} s^{-1}$; $E_i \approx \Delta H^\circ_{300}$.

TABLE III: Estimated (High Pressure) Rate Constants for C-C Rupture in Hydrocarbon Radicals

reaction	$\log A_{\text{rev}}, \text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$	$E_{\text{rev}}, \text{kcal mol}^{-1}$	ref	$\log A_i, s^{-1}$	$E_i, \text{kcal mol}^{-1}$	$\Delta H^\circ_{300}, \text{kcal mol}^{-1}$
A. Production of Saturated Radicals ^a						
$\text{CC}\dot{\text{C}} \rightarrow \text{CH}_3 + \text{C}=\text{C}$	11.5	7.7	24	13.0	32.9	26.4
$\text{CC}=\dot{\text{C}} \rightarrow \text{CH}_3 + \text{C}\equiv\text{C}$	11.8	7.7	24	13.1	33.4	26.2
$\text{CC}\dot{\text{C}}\text{C} \rightarrow \text{CH}_3 + \text{C}=\text{CC}$	10.9	7.1	24	12.8	32.3	26.7
$\text{CC}=\dot{\text{C}}\text{C} \rightarrow \text{CH}_3 + \text{C}\equiv\text{CC}$	11.7	8.8	24	13.7	31.3	23.4
$\text{CCCC}\dot{\text{C}} \rightarrow \text{C}_2\text{H}_5 + \text{C}=\text{C}$	11.2	7.3	24	12.7	27.8	22.4
B. Production of Vinylic Radicals ^b						
$\text{C}=\text{CC}=\dot{\text{C}} \rightarrow \text{C}=\dot{\text{C}} + \text{C}\equiv\text{C}$	11.8	4.7	8	14.0	43.9	40.4
$\text{C}=\text{CC}=\dot{\text{C}} \rightarrow \text{C}=\dot{\text{C}} + \text{C}\equiv\text{C}$	11.2	3.6	25	13.4	42.8	40.4
$\text{C}=\text{CCCC}=\dot{\text{C}} \rightarrow \text{C}=\dot{\text{C}} + \text{C}=\text{CC}=\text{C}$	11.5	3	8	13.3	47.9	46.3
$\text{C}=\text{CC}=\text{CC}=\dot{\text{C}} \rightarrow \text{C}=\text{CC}=\dot{\text{C}} + \text{C}\equiv\text{C}$	11.5	3.7	25	13.5	43.1	40.9
C. Production of Ethynyl ^c						
$\text{C}=\text{CC}=\dot{\text{C}} \rightarrow \text{C}\equiv\text{C} + \text{C}\equiv\text{C}$	13.5	0	1	14.5	57.2	58.5

^a Suggested values: $A_i \approx 2 \times 10^{13} s^{-1}$; $E_i \approx \Delta H^\circ_{300} + 7 \text{ kcal mol}^{-1}$. ^b Suggested values: $A_i \approx 4 \times 10^{13} s^{-1}$; $E_i \approx \Delta H^\circ_{300} + 2 \text{ kcal mol}^{-1}$. ^c Suggested values: $A_i \approx 3 \times 10^{14} s^{-1}$; $E_i \approx \Delta H^\circ_{300} - 1 \text{ kcal mol}^{-1}$.

TABLE IV: Estimated (High Pressure) Rate Constants for C-H Rupture in Hydrocarbon Radicals^a

reaction [ref]	$\log A_{\text{rev}}, \text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$	$E_{\text{rev}}, \text{kcal mol}^{-1}$	$\log A_i, s^{-1}$	$\log [A_i/\#H], s^{-1}$	$E_i, \text{kcal mol}^{-1}$	$\Delta H^\circ_{300}, \text{kcal mol}^{-1}$
$\text{C}_2\text{H}_5 \rightarrow \text{C}=\text{C} + \text{H}$, [22]	13.6	2.6	13.7	13.2	40.9	38.7
$\text{CC}\dot{\text{C}} \rightarrow \text{CC}=\text{C} + \text{H}$, [22]	12.9	2.9	13.1	12.8	38.5	36.0
$\text{CCC}\dot{\text{C}} \rightarrow \text{CCC}=\text{C} + \text{H}$, [22]	12.9	2.9	13.0	12.7	38.3	36.1
$\text{C}=\text{C}\dot{\text{C}} \rightarrow \text{C}=\text{C}=\text{C} + \text{H}$, [24]	12.6	2.7	13.1	13.1	61.3	58.9
$\text{CC}\dot{\text{C}} \rightarrow \text{CC}=\text{C} + \text{H}$, [22]	12.9	1.2	13.6	12.8	39.6	38.8
$\text{CC}\dot{\text{C}}\text{C} \rightarrow \text{CCC}=\text{C} + \text{H}$, [22]	12.9	1.2	13.2	12.7	39.6	39.1
$\text{CC}\dot{\text{C}}\text{C} \rightarrow \text{CC}=\text{CC} + \text{H}$, [22]	12.7	1.2	12.5	12.2	36.9	36.4
	13.6	4.0	13.5	13.2	28.5	24.9
$\text{C}=\text{C}\dot{\text{C}} \rightarrow \text{C}\equiv\text{C} + \text{H}$, [24]	12.7	2.4	12.5	12.2	38.3	35.9
$\text{CC}=\dot{\text{C}} \rightarrow \text{CC}=\text{C} + \text{H}$, [24]	12.8	3.1	12.6	12.6	36.5	33.6

^a Suggested values: $A_i/\#H \approx 6 \times 10^{12} s^{-1}$; $E_i \approx \Delta H^\circ_{300} + 1 \text{ kcal mol}^{-1}$ (secondary radicals) and $E_i \approx \Delta H^\circ_{300} + 2.5 \text{ kcal mol}^{-1}$ (primary radicals).

the resonant structure of the product inhibits rotation about the C-C bond.²³ Note in all cases one can get a good estimate of the barrier by the simple relation $E_i \approx \Delta H^\circ_{300} - 1 \text{ kcal mol}^{-1}$

(ΔH°_{300} based upon a standard state of 1 atm).

Table II summarizes the results from C-H dissociation in stable molecules. The recombination rate constants were assumed to be $k = 1 \times 10^{14} \text{ cm}^3 \text{mol}^{-1} \text{s}^{-1}$ with no temperature dependence.²² Again, the generic A factor is probably within a factor of 2 of the values for any specific reaction. As with C-C rupture, one must account for formation of resonantly stabilized fragments by using a lower A factor.

The situation becomes more complex when considering dissociation of radicals since there is more uncertainty with respect

(23) Benson, S. W. "Thermochemical Kinetics", 2nd ed.; Wiley: New York, 1976; p 54.

(24) Kerr, J. A., Moss, S. J., Eds. "Handbook of Bimolecular and Ter-molecular Gas Reactions"; CRC Press: Boca Raton, FL, 1981; Vol. II.

(25) Cole, J. A., M. S. Thesis, Massachusetts Institute of Technology, 1982.

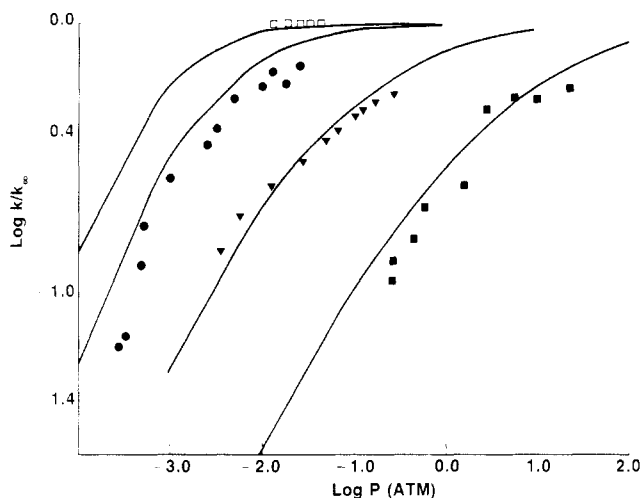


Figure 1. Comparison of observed and predicted falloff curves for $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$. Observed values (symbols) taken from Warnatz;¹ $\square$, $T = 298$ K; $\bullet$, $T = 520$ K; $\blacktriangledown$, $T = 913$ K; $\blacksquare$, $T = 1350$ K. Solid lines are predicted values for these same temperatures.

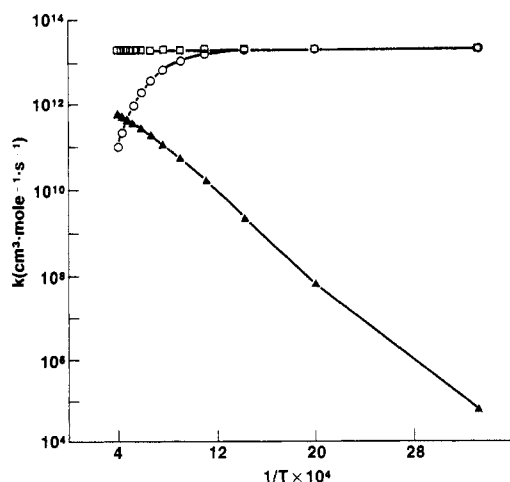


Figure 2. Effect of temperature upon $\text{CH}_3 + \text{CH}_3$ recombination at 1 atm. Upper line ($\square$) is the limiting high-pressure rate constant for formation of C_2H_6 ;²² $\circ$ = predicted apparent rate constant for $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$; $\blacktriangle$ = predicted apparent rate constant for $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_5 + \text{H}$.

to the rate constants for the reverse reactions, addition to unsaturates. The values for C–C rupture in Table III were obtained by using specifically measured addition rate constants with the appropriate equilibrium constant. As one might expect, the resulting spread of forward rate constants is somewhat larger here, with uncertainties probably closer to a factor of 3 for the A factor.

Table IV summarizes the results for a series of C–H ruptures in radicals. Again one finds appreciable scatter, with uncertainties in the A factor on the order of a factor of 3. Note there appears to be no clear correlation between the A factor and radical structure. As before one can obtain a good estimate of E_i from ΔH°_{300} . The larger correction term for primary radicals simply mirrors the larger barrier for the reverse reaction there. (In this context, addition to benzene seems to be a special case.)

Using suggested values for the dissociation parameters of the energized complexes that are listed in Tables I–IV, one has essentially all the information required to implement the approach outlined in the previous section. The number of vibrational degrees of freedom is simply $3N - 6$, where N is the number of atoms in the adduct. The geometric mean frequency ν can be estimated quite reliably from published data on stable molecules.²⁶ The

values for radicals will generally be quite close to that of the stable species with one additional hydrogen atom. For example, CH_3 ($\nu = 1798 \text{ cm}^{-1}$ ²⁷) is 159 cm^{-1} lower than CH_4 , while the calculated value for vinyl ($\nu = 1561 \text{ cm}^{-1}$ ²⁸) is only 27 cm^{-1} lower than ethylene. For larger radicals, the differences should be even smaller.

Applications

1. Falloff Curves for Recombination Reactions. As indicated in the Introduction, the substantial shift of the falloff curve of methyl radical recombination toward higher pressures with increasing temperature is a manifestation of the rapid increase in the internal energy of the collision complex with temperatures. Warnatz¹ has used QRRK theory to compute the falloff curves based upon the dissociative reaction. The same results should be obtained by considering the recombination path because of microscopic reversibility. Such a calculation was used to verify the validity of the generic rate constants in Tables I and II. The mechanism (eq 7–9) fits scheme A with k_{stab} (the apparent rate constant for stabilization) and k_{rxn} (the apparent rate constant for “reaction”, i.e., producing $\text{C}_2\text{H}_5 + \text{H}$) computed from eq 10 and 11. Following Tables I and II, $A_{-1} = 1 \times 10^{17} \text{ s}^{-1}$, $E_{-1} = 88.8 \text{ kcal mol}^{-1}$, $A_2 = 1.8 \times 10^{16} \text{ s}^{-1}$, $E_2 = 98.2 \text{ kcal mol}^{-1}$. Argon was the third body with $-\langle \Delta E \rangle = 630 \text{ cal mol}^{-1}$,²⁰ ν was taken to be 1509 cm^{-1} .²⁶ The apparent rate constants k_{stab} and k_{rxn} are computed for a range of pressures at a given temperature and the process repeated at another temperature. The computed index of falloff, $k/k_\infty (= k_{\text{stab}}/k_1)$, is compared in Figure 1 to the data compiled by Warnatz. The agreement is most gratifying; a particularly satisfactory aspect is that this fit was obtained by using physically reasonable parameters which were not adjusted.

Figure 2 shows the predicted behavior of k_{stab} and k_{rxn} as a function of temperature at 1 atm. Here one can see the dramatic differences between application of the present approach and use of the assumption that the recombination is always at the high-pressure limit (the upper curve in Figure 2). Note that the model predicts adequate energy buildup in the complex to start falling off from the high-pressure limit by $\sim 900 \text{ K}$ at 1 atm. By 2000 K the rate constant of the recombination channel has dropped by a factor of 50. The other manifestation of the rapid increase in energy with temperature is the onset of the reaction channel. These calculations predict this channel will dominate above 2100 K at 1 atm.

Several groups have reported high-temperature values for k_{rxn} ; unfortunately there is substantial disagreement as to the correct result. Roth and Just⁹ reported $k_{\text{rxn}} = 8 \times 10^{14} \exp(-26.5 \text{ kcal/RT}) \text{ cm}^3/\text{mol}^{-1} \text{ s}^{-1}$ for $T = 1650$ – 2100 K whereas Kiefer and Budach¹⁰ reported $k_{\text{rxn}} = 8 \times 10^{11} \exp(-13.0 \text{ kcal/RT})$ over a comparable range. At high temperatures, the present calculations show that k_{rxn} is quite insensitive to pressure, with $k_{\text{rxn}} \cong 3.6 \times 10^{12} \exp(-8.82 \text{ kcal/RT})$ for $T = 1500$ – 2500 K and $P = 1 \text{ atm}$. Figure 3 compares these results to the data and to a very recent RRKM calculation by Larson and Golden.¹¹ The two calculated curves are reasonably consistent, differing by a factor of 2 or less between 1000 and 2000 K .

Given the ease of applying QRRK to this case where only the generic rate parameters were used, such agreement with the more elaborate RRKM approach is quite satisfying. It would appear that one can indeed develop reliable estimates of the pressure and temperature dependences of both recombination and reaction channels via the simpler approach outlined in this paper. The present results clearly reinforce the message of Larson and Golden that the “reaction channel” might play a significant role in this particular reaction.

(26) Shimanouchi, T. “Tables of Molecular Vibration Frequencies, Consolidated Vol. I”; *Natl. Stand. Ref. Data Ser. (U.S. Natl. Bur. Stand.)* **1972**, NSRDS-NBS 39.

(27) “JANAF Thermochemical Tables”, 2nd ed.; *Natl. Stand. ref. Data Ser. (U.S. Natl. Bur. Stand.)* **1971**, NSRDS-NBS 37.

(28) Dupuis, M.; Wendoloski, J. J. *J. Chem. Phys.* **1984**, *80*, 5696.

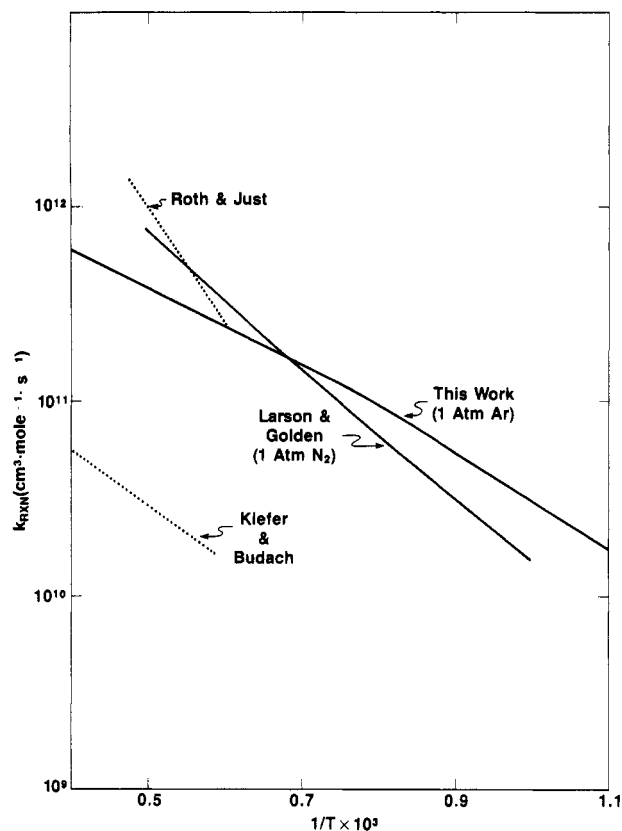


Figure 3. Comparison of measured and predicted values for the rate constant of $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6 + \text{H}$. Dotted lines are measured: Roth and Just⁹ and Kiefer and Budach.¹⁰ Solid lines are predicted: this work (QRRK with generic rate constants) and Larson and Golden¹¹ (RRKM).

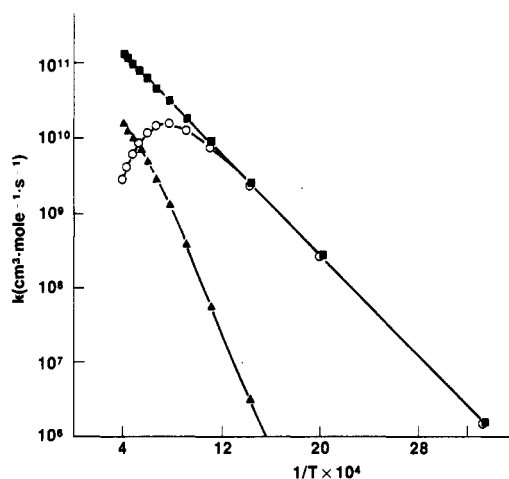


Figure 4. Predicted effect of temperature upon $\text{CH}_3 + \text{C}_2\text{H}_2$ addition at 1 atm. Upper line (■) is the limiting high-pressure rate constant;²⁴ ○ = predicted apparent rate constant for $\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{CC}=\text{C}$; ▲ = predicted apparent rate constant for $\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{CC}\equiv\text{C} + \text{H}$.

2. *Radical Addition to Unsaturated Hydrocarbons.* One can examine the role of different barrier heights for the C–H fission channel by considering the following reactions:

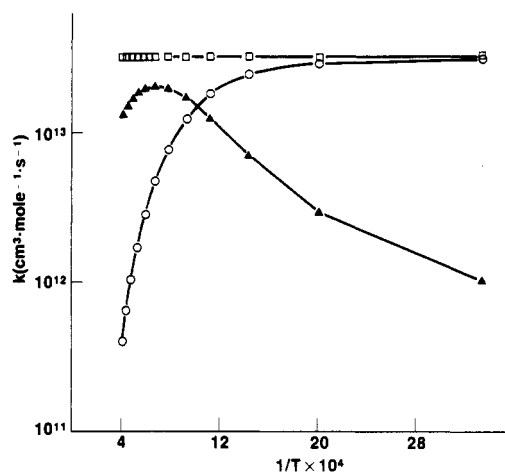
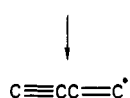
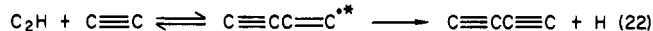
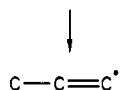
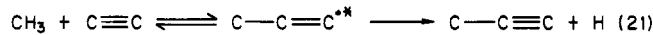


Figure 5. Predicted effect of temperature upon $\text{C}_2\text{H} + \text{C}_2\text{H}_2$ addition at 1 atm. Upper line (□) is the limiting high-pressure rate constant [1]; ○ = predicted apparent rate constant for $\text{C}_2\text{H} + \text{C}_2\text{H}_2 \rightarrow \text{C}\equiv\text{CC}=\text{C}$; ▲ = predicted apparent rate constant for $\text{C}_2\text{H} + \text{C}_2\text{H}_2 \rightarrow \text{C}\equiv\text{CC}\equiv\text{C} + \text{H}$.

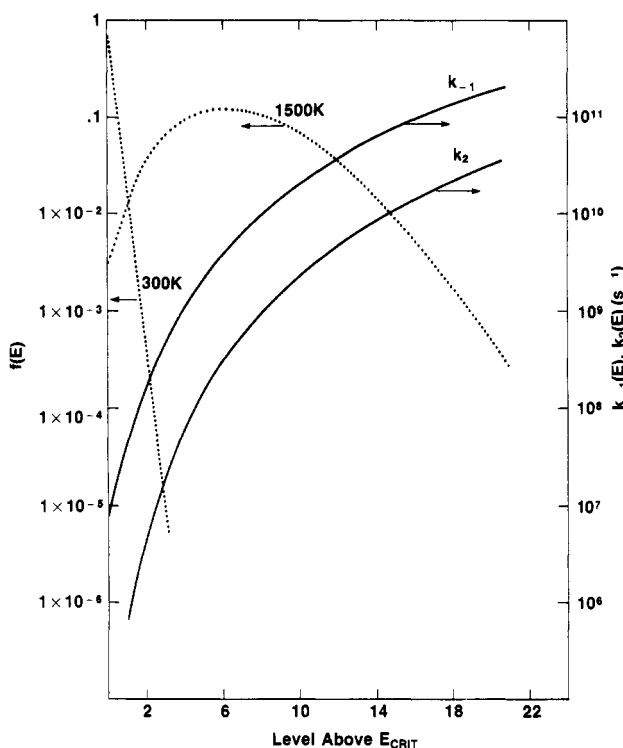


Figure 6. Details for the calculation of $\text{CH}_3 + \text{C}_2\text{H}_2$. Dotted lines are the chemical activation distribution functions at 300 and 1500 K for various levels above E_{crit} (step size = 1300 cm^{-1}). Solid lines are QRRK rate constants for k_{-1} ($\text{CC}=\text{C}^* \rightarrow \text{CH}_3 + \text{C}_2\text{H}_2$) and k_2 ($\text{CC}=\text{C}^* \rightarrow \text{CC}\equiv\text{C} + \text{H}$).

For CH_3 addition, the barrier to H elimination is $\sim 2 \text{ kcal mol}^{-1}$ higher than that required for the complex to dissociate back to reactants. For C_2H addition, the corresponding C–H rupture channel is $\sim 21 \text{ kcal mol}^{-1}$ lower in energy than the C–C route. Calculations for these cases over a range of temperatures at 1 atm are shown in Figures 4 and 5. To simulate a hydrocarbon environment, CH_4 was chosen as the bath gas with $-(\Delta E) \cong 2.1 \text{ kcal mol}^{-1}$. Both of these calculations reinforce the necessity of explicitly considering the (complicating) role of temperature.

It is immediately obvious that use of the usual assumption that addition reactions can be considered to be near their high-pressure limiting behavior is not valid at even moderately elevated temperatures. For CH_3 addition (Figure 4), note not only that the apparent rate constant for formation of the stabilized adduct drops rapidly at higher temperatures, but also that the “new” channel

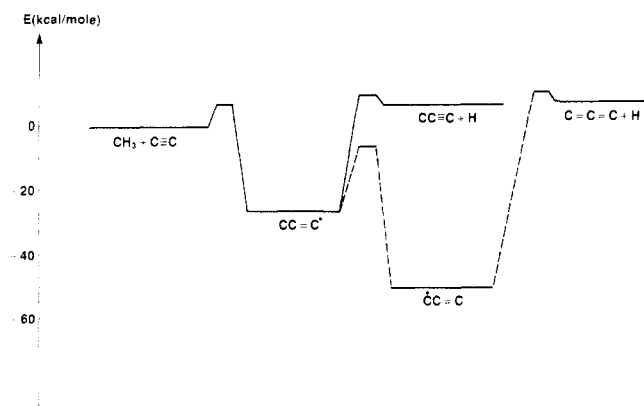
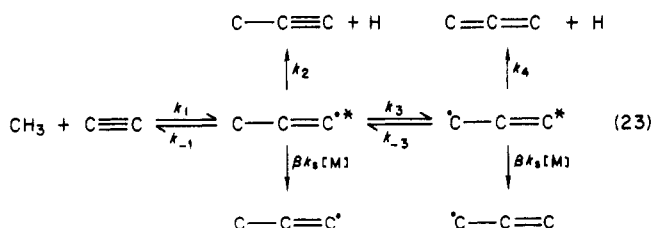


Figure 7. Potential energy diagram for $\text{CH}_3 + \text{C}_2\text{H}_2$ addition. Dotted lines represent a 20 kcal mol^{-1} barrier for the 1,3 H shift with subsequent C-H rupture.

producing H atoms dominates there. For C_2H addition (Figure 5), the H atom elimination channel becomes important at much lower temperatures. This is to be expected given the low barrier for this channel with its correspondingly rapid dissociation rates. The drop in the H elimination channel at the highest temperatures is a manifestation of the increasing importance of dissociation of the complex back to reactants under these conditions. Although there is a higher barrier for this reaction, the higher A factor begins to dominate at high temperatures.

The rapid increase in H atom production with temperature in Figure 4 can be understood in terms of the increase in energy of the complex as shown in Figure 6. There the distribution function $f(E)$ is compared at 300 and 1500 K; the variations of the rate constants for both dissociation channels with energy are also shown. At low temperature the rate of dissociation via k_2 (C-H rupture) or k_{-1} (C-C rupture) is too slow relative to stabilization ($\beta k_s[M] \sim 8 \times 10^9 \text{ s}^{-1}$ at 1 atm) for all levels populated; thus virtually all adducts are collisionally stabilized and one is in the high-pressure limit. At higher temperatures, the population shifts to higher levels and dissociation competes with stabilization.

3. Isomerization of the Energized Complex. Two particularly interesting possibilities for isomerization at 1,3 hydrogen shifts and cyclization of larger radicals. The former are generally considered to be unlikely in thermal hydrocarbon systems, since the strained ring intermediate should result in a relatively large activation energy. Tardy²⁹ has reported barriers on the order of 30 kcal mol^{-1} . In the context of the present work, however, the energized complex could easily have energies comparable to this and such a channel might be feasible. Such a possibility was considered for one of the addition reactions previously discussed



thus opening up the possibility of forming allyl radicals and/or allene. In light of Tardy's results, a barrier of 20 kcal mol^{-1} was assigned as a lower limit for the 1,3 shift. A potential energy diagram for this case is shown in Figure 7. This case has particularly favorable energetics for the 1,3 shift since the vinylic C-H bond in ~ 24 kcal mol^{-1} stronger than the allylic C-H bond.³⁰ The A factors for both k_3 and k_{-3} were chosen to be $2 \times 10^{11} \text{ s}^{-1}$, similar to that for intramolecular H transfers involving less strained transition states;⁸ other rate constants were assigned as indicated

(29) Tardy, D. C. *Int. J. Chem. Kinet.* **1974**, *6*, 291.

(30) McMillen, D. F.; Golden, D. M. *Annu. Rev. Phys. Chem.* **1982**, *33*, 493.

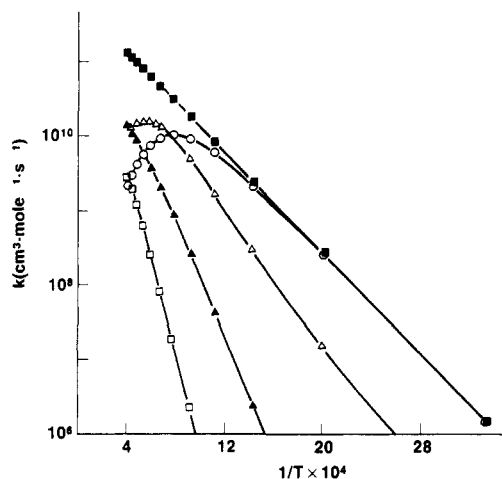
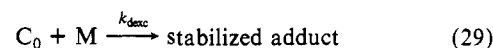
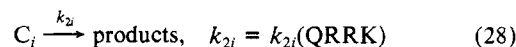
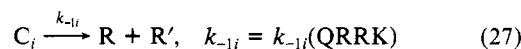
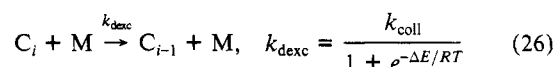
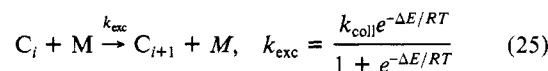
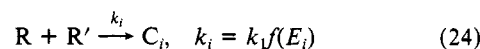


Figure 8. Predicted effect of including 1,3 H shift with 20 kcal mol^{-1} barrier for $\text{CH}_3 + \text{C}_2\text{H}_2$ addition at 1 atm. Upper line (■) is the limiting high-pressure rate constant;²⁴ ○ = predicted apparent rate constant for $\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{CC}\equiv\text{C}$; Δ = predicted for $\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{C}\equiv\text{C}-\text{C}$; ▲ = predicted for $\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{CC}\equiv\text{C} + \text{H}$; □ = predicted for $\text{CH}_3 + \text{C}_2\text{H}_2 \rightarrow \text{C}\equiv\text{C}=\text{C} + \text{H}$.

in Tables III and IV. The results at 1 atm are shown in Figure 8. The allyl channel can actually dominate this system at higher temperature. This increase comes at the expense of the C-C=C channel with less of an effect upon the CC≡C + H route. (Very little allene is formed since this barrier is so high.) Thus at least in this case one might well need to consider the possibility of 1,3 H shifts in addition reactions.

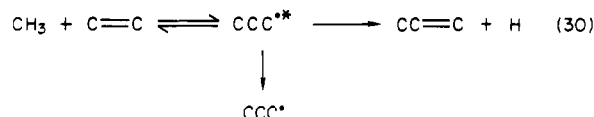
Some particularly interesting kinetic possibilities arise from consideration of cyclization of the energized complex to form aromatic precursors. This is now being treated in detail for butadiene pyrolysis³¹ and for molecular weight growth in flames.³²

4. Master Equation Results. The accuracy of treating collisional stabilization simply in terms of a deactivation efficiency β obtained from the magnitude of the energy transferred per collision was explicitly checked in two limiting cases. The following scheme was used:



where $f(E_i)$ = fraction of C at energy E_i (eq 12), k_1 = (overall) rate constant for adduct formation, k_{coll} = Lennard-Jones collisional rate constant, ΔE = (equal) energy between adjacent levels, and the dissociation rates k_{-1i} and k_{2i} are obtained with the QRRK expression (eq 14). This simple one-step energy-transfer scheme assumes stabilization only when molecules in the lowest level (C_0) with energy E_{crit} undergo deactivation.

The first reaction considered was



(31) Dean, A. M.; Woodin, R. L., manuscript in preparation.

(32) Dean, A. M.; Westmoreland, P. R., manuscript in preparation.

TABLE V: Effect of More Realistic Energy-Transfer Model for $\text{CH}_3 + \text{C}_2\text{H}_4$

		P		
		20 torr	1 atm	10 atm
T = 300 K	master eq			
	$[\text{CC}\dot{\text{C}}]/[\text{CC}=\text{C}]$	3.8×10^4	1.4×10^6	1.4×10^7
	QRRK, eq 10 and 11			
	$k_{\text{stab}}/k_{\text{rxn}}$	4.6×10^4	1.7×10^6	1.7×10^7
T = 1500 K	master eq			
	$[\text{CC}\dot{\text{C}}]/[\text{CC}=\text{C}]$	0.046	0.78	3.73
	QRRK, eq 10 and 11			
	$k_{\text{stab}}/k_{\text{rxn}}$	0.25	3.11	15.2
	$[\text{CC}\dot{\text{C}}]/[\text{CC}=\text{C}]$ ($k_{\text{stab}}/k_{\text{rxn}}$)	0.18	0.25	0.24

The barrier to C–H rupture was $5.5 \text{ kcal mol}^{-1}$ higher than that for C–C rupture so that one might expect complications as discussed earlier. ΔE was set at $1.85 \text{ kcal mol}^{-1}$, one-half the value of ν in the usual β -type calculations for this system. The QRRK rate constants for these “half-steps” were estimated via interpolation. The lower ΔE was used to more accurately simulate typical collisions with hydrocarbons where $\Delta E \approx 2 \text{ kcal mol}^{-1}$.

Results of the numerical integrations revealed the system rapidly approached a steady state. Table V compares these results to those obtained by using the modified strong collision model. In general, comparisons are most encouraging. At low temperatures only a slight bias ($\sim 20\%$) is introduced. Slightly less stabilization is produced via the master equation since the strong deactivation collision assumption in eq 10 and 11 precludes the possibility of the complex dissociating via k_{-1} on its way toward stabilization. As one might expect, the bias is larger at elevated temperatures; this is a direct consequence of the larger number of levels populated—one has much more chance of a cascade of collisions and the assumption of a single deactivating collision which removes all the excess energy is much less tenable. One encouraging observation is that the extent of the error is quite insensitive to pressure. The effect of the strong collision assumption is to *overestimate* the amount of stabilized complex relative to H elimination. In other words, the predictions with the energized complex theory can be viewed as a *lower limit* on the amount of complex reacting via the “new” pathway, since most of the H-atom eliminations after CH_3 additions will have energetics similar to the case treated here.

As discussed earlier, one might expect severe difficulties when considering systems with very low energy exit channels, as in C_2H addition to unsaturates. Master equation calculations were done for $\text{C}_2\text{H} + \text{C}_2\text{H}_2$ with $\Delta E = 2.57 \text{ kcal mol}^{-1}$. (This ΔE corresponds to the assigned geometric mean frequency for $\text{C}\equiv\text{CC}=\dot{\text{C}}$.) The results are compared to those obtained with the modified strong collision assumption in Table VI. At 300 K, very little difference is seen. The fact that the master equation suggests more stabilization is probably due to use of a somewhat larger ΔE than used for the modified strong collision approach where ΔE was $2.1 \text{ kcal mol}^{-1}$.

At 1500 K, the master equation solution yields appreciably less stabilization. This is qualitatively consistent with the large number of levels populated; many collisions are required to remove enough energy for these adducts to be stabilized. Since the barrier to form $\text{C}_4\text{H}_2 + \text{H}$ is so low (21 kcal mol^{-1} lower than the barrier to $\text{C}_2\text{H} + \text{C}_2\text{H}_2$), the dissociation rate for this channel will still be very fast even after a molecule has lost some energy via collisions. At lower pressure, with longer times between collisions during which dissociation could occur, one might expect to see the largest differences between the two methods; this trend is evident in Table VI. As with CH_3 addition, the comparison reveals the tendency of the modified strong collision model to overestimate the extent of stabilization. Although the data in Table VI show substantial differences at 1500 K and 20 torr, it is worthwhile to note that both approaches indicate virtually complete formation of the

TABLE VI: Effect of More Realistic Energy-Transfer Model for $\text{C}_2\text{H} + \text{C}_2\text{H}_2$

		P		
		20 torr	1 atm	10 atm
T = 300 K	master eq			
	$[\text{C}\equiv\text{CC}=\dot{\text{C}}]/[\text{C}\equiv\text{CC}=\text{C}]$	1.09	38.9	388.0
	QRRK, eq 10 and 11			
	$k_{\text{stab}}/k_{\text{rxn}}$	0.82	30.2	301.0
T = 1500 K	master eq			
	$[\text{C}\equiv\text{CC}=\dot{\text{C}}]/[\text{C}\equiv\text{CC}=\text{C}]$ ($k_{\text{stab}}/k_{\text{rxn}}$)	1.33	1.29	1.29
	QRRK, eq 10 and 11			
	$k_{\text{stab}}/k_{\text{rxn}}$	1.44×10^{-4}	0.0278	0.396
	$[\text{C}\equiv\text{CC}=\dot{\text{C}}]/[\text{C}\equiv\text{CC}=\text{C}]$ ($k_{\text{stab}}/k_{\text{rxn}}$)	6.97×10^{-3}	0.228	1.68
	$[\text{C}\equiv\text{CC}=\dot{\text{C}}]/[\text{C}\equiv\text{CC}=\text{C}]$ ($k_{\text{stab}}/k_{\text{rxn}}$)	0.021	0.122	0.236

dissociation products under these conditions.

A comparison of Tables V and VI reveals the importance of the energetics of the exit channel. For example, at 300 K and 20 torr, note that the tendency of $\text{C}-\text{C}\dot{\text{C}}^*$ to be stabilized is $\sim 10^4$ larger than $\text{C}\equiv\text{CC}=\dot{\text{C}}^*$; this is primarily due to the availability of the low energy $\text{C}\equiv\text{CC}=\text{C} + \text{H}$ channel for the latter case. At higher temperatures the spread of the distribution function tends to minimize this difference, but note that at 1500 K and 20 torr, the likelihood of stabilization of $\text{CC}\dot{\text{C}}^*$ is still ~ 300 times larger than $\text{C}\equiv\text{CC}=\dot{\text{C}}^*$.

In light of the generally encouraging comparisons between the strong collision calculations and the master equation results, the approach outlined in this paper represents a relatively simple method to estimate the effects of pressure and temperature on both recombination and addition rate constants. Of course, the ease of generating those estimates is offset by the uncertainties introduced via use of generic rate constants and use of a modified strong collision assumption within a QRRK framework. In spite of these limitations, development of a detailed kinetic model using these methods to estimate addition and recombination rate constants should provide a much better description of many high-temperature processes. Applications of sensitivity analysis³³ to the results of this initial mechanism would then indicate which specific reactions warrant further attention. Hopefully, this number will be sufficiently small to allow more careful analysis; here one would have the incentive to apply RRKM methods and consider master equation treatments. In this light, the approach outlined above makes it much easier to get started on the road which leads to a better understanding of addition and recombination reactions in complex systems.

Summary

The approach developed in this paper describes a relatively straightforward means to estimate pressure and temperature effects on reactions which proceed through formation of an energized complex. The specific applications considered indicate the extremely complex behavior that such reactions can exhibit. Just as it has become apparent in the last few years that many important recombination reactions are in their falloff region at higher temperatures, this paper now shows that similar considerations apply to addition reactions. Even more important, the same energetics which shift the falloff curves allow entirely new reaction pathways to open up. This work has shown how it can be quite common for either addition or recombination reactions involving hydrocarbon radicals to result in production of H atoms. Such H atom production can have dramatic kinetic consequences, as it is generally much more reactive than the radical(s) from which it was created. Other implications relating to the rapid formation

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of aromatics in both pyrolysis and oxidation environments are being actively pursued.

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Measurement of Internal Energies by Hot Ultraviolet Absorption Spectroscopy: Spectra of Excited Azulene Molecules

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The gas-phase absorption spectrum of azulene in the wavelength range $\lambda = 220$ – 300 nm has been measured in shock waves between 600 and 1750 K. Combined with the room-temperature spectrum, a dependence of the absorption coefficients $\epsilon(\lambda)$ on the average thermal energy $\langle E \rangle$ over the range 10–420 kJ mol⁻¹ is established. The "canonically hot" UV spectra are compared with "microcanonically hot" UV spectra obtained by laser excitation of azulene at 337 and 308 nm, corresponding to excitation energies of 367 and 400 kJ mol⁻¹. "Microcanonically hot" and "canonically hot" absorption spectra agree perfectly when the average internal energies $\langle E \rangle$ of the ensembles are the same. This experimental observation is rationalized. Various applications of hot UV absorption spectroscopy in chemical kinetics are reviewed.

Introduction

Vibrationally highly excited molecular states play an important role in a vast class of chemical reactions. For this reason, a spectroscopic detection of these states is of great interest. There are various spectroscopic techniques which can be used for a time-resolved observation of excited-state populations. Fluorescence or multiphoton ionization is particularly sensitive. State-resolved absorption, fluorescence excitation, or stimulated emission is useful whenever selected individual states can be separated. There exist, however, situations where the density of states is so large that one cannot realize a resolution of single states or where neither fluorescence nor multiphoton ionization signals can be detected. Such situations are frequently met with vibrationally highly excited polyatomic molecules in the electronic ground state. In this case, "hot UV absorption spectroscopy" can be used for the time-resolved observation of excited-state populations. In our laboratory we have frequently used this type of spectroscopy for the observation of the dynamics of intramolecular processes, such as isomerization and bond fission under isolated molecule conditions, as well as of collisional intermolecular energy transfer.

Spectroscopists have not too intensely looked into "hot UV absorption" because of the lack of state resolution. However, this type of spectroscopy may provide a direct access to the average internal energy of the excited molecules, more or less independent of the energy distributions. For this reason, apart from just indicating the presence of excited species, this spectroscopy can serve as an "internal energy thermometer". In order to prove this statement, in the present work we report a key experiment with excited azulene molecules. Before doing this, we briefly describe some applications of hot UV absorption spectroscopy.

Well-known effects of temperature on absorption spectra are the broadening of continuous spectra, like in Cl₂ near 330 nm,¹ or the increasing number and overlap of hot bands in spectra with rovibrational structure, like in CS₂ near 200 nm.^{2,3} The phenomenon is well documented from the low temperatures of jet-cooled molecules up to high temperatures realized in shock waves.

Apart from Cl₂ and CS₂, one may cite the examples N₂O,⁴ O₃,⁵ CF₃I,⁶ benzene,⁷ toluene,^{7,8} cycloheptatrienes,⁸ and others. Often there are wavelengths where the absorption coefficient ϵ depends strongly on the temperature, such that it can serve as a spectroscopic thermometer (see e.g. ref 9–11). If the absolute value of ϵ is not sufficient, its temperature-dependent wavelength dependence can also be evaluated. The response of this thermometer is very fast such as demonstrated recently in picosecond experiments by Kaiser and co-workers.¹¹

Theoretical predictions of the temperature dependence of UV absorption spectra over large temperature ranges are still very limited. The Franck–Condon principle, applied to suitable potential energy curves, has allowed for an accurate analysis of the temperature dependence of continuous spectra of halogens in the visible or near-UV.^{1,12} With increasing temperature, the absorption decreases at the center and increases at the wings of the nearly Gaussian profiles of the continuous absorption spectrum. By employing several approximations, the Sulzer–Wieland treatment¹³ satisfactorily describes this behavior by a simple formula. It may appear somewhat surprising that the continua of polyatomic molecules, in spite of their more complex excited-state geometries, often show nearly the same behavior^{8,14} and, to

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